

1	(a)		for a body in (rotational) equilibrium, sum of/total clockwise moments <u>about a point</u> = sum of/total anticlockwise moments <u>about the same point</u>	B1
	(b)	(i)	1. (Taking moment about P,) $3.4 \times 10^4 \times 5 = T \times 12$ $T = 14\,200\text{ N}$ <i>correct substitution for all terms</i>	M1 A0
			2. moment of vertical rope = $T \cos 24^\circ \times 2.5$ (Taking moment about P,) $T \cos 24^\circ \times 2.5 = F \times 8$ $T = 4050\text{ N}$ <i>correct substitution for all terms</i>	C1 M1 A0
		(ii)	Assume that mass of beam is negligible / light $R_Y = 14200 + 4050 \cos 24.0^\circ = 17900\text{ N}$ $R_X = 4050 \sin 24.0^\circ = 1647\text{ N}$ $\alpha = \tan^{-1} \frac{R_X}{R_Y}$ $\alpha = 5.26^\circ, R = 18000\text{ N}$	C1 C1 A1
	(b)		trailer accelerates forward, require horizontal component of force vertical component of force remains the same, so magnitude of F increases resultant of the components at an angle to the vertical towards the car	B1 B1 B1
				[Total: 10]